

Gabriel Chen

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Education

ScB in Mathematics and Computer Science, BA in Linguistics

Expected May 2028

Brown University, Providence, Rhode Island

- GPA: 4.0/4.0
- **Coursework:** Multivariable Calculus, Statistics, Linear Algebra, Functional Programming

Experience

Head Teaching Assistant, MIT Beaver Works Summer Institute

July - August 2024, 2025

- Led and organized team of 10 Teaching Assistants to develop course materials, increase student engagement, and teach software design principles
- Assisted students with debugging Python code for AI-driven game projects
- Taught fundamental game design principles and Agile software design methodologies
- Delivered presentation on linguistics and its applications in gaming
- Organized and led after-camp social activities to foster student engagement and collaboration
- Developed a 2048 AI for a competition against student-developed AI's

Ground Station Software-Defined-Radio Design Lead, Brown Space Engineering

September 2024 – Present

- Develop and maintain software for receiving and processing telemetry data from the PVDX satellite using platforms such as SatNOGS.
- Helped integrate hardware solutions (HackRF One) into telemetry systems to optimize data capture.

Teacher and Curriculum Developer, Brown IgniteCS

September 2024 – Present

- Developed coding lesson plans tailored to middle school students at Nathanael Greene Middle School, fostering engagement, building foundational coding skills, and emphasizing data structures and algorithms.

Invitational Round Competitor and Organizer, North American Computational Linguistics Olympiad (NACLO)

2022 - 2024

- Organized the NACLO competition at Memphis University School
- Conducted workshops to prepare participants for syntax, semantics, and pragmatics problems
- Competed in Invitational Round after placing in top 10% of Open Round

Projects

Game-Based Study of Human and AI Decision-Making

- Developed a Serious Game designed to assess how human decision-making aligns with AI across various ethically challenging scenarios.
- Tools Used: Python, Unity

English Compound Word Deconstructor

- Trained a neural network on the LADEC (Large Database of English Compounds) to reliably split English compound words into their component words.
- Tools Used: Python

Proficiencies

Programming Languages: C++, C, C#, Java, Python, Racket, Unity, React, Pyret

Languages: English, Mandarin, Latin